JC2 H2 Biology Preliminary Examination Answer Key

Section A

Question 1

(a) Identify this stage in the cell cycle and the structure labeled A as shown in Figure 4.1.

Stage: telophase/cytokinesis **A:** cell membrane. **[1]**

(b) Suggest why this stage is the least frequently observed stage in mitosis.

This phase is the shortest and cell goes into interphase almost immediately,
Difficult to differentiate between cells in interphase/cytokinesis **[1]**

(c) Explain why DNA amount and chromosome numbers remain constant during the two anaphases in Figure 4.2?

DNA content and chromosome number is with reference to a **nucleus within one cell** (1m). During anaphase, the chromosomes are moving towards the pole, (1m) the nucleus is not formed yet and the cell has not divided into two individual cells (1m) thus DNA content and chromosome number will not change / will remain constant. **Max [2]**

(d) List two ways in which prophase of *meiosis I* would differ from prophase in *mitosis*.

- In meiosis I **homologous chromosome** would pair up; in metaphase of mitosis, homologous chromosome will not pair up
- In meiosis I, there will be crossing over between **non-sister** chromatids of homologous chromosomes while in mitosis, no such event occurs. **Max [2]**

(e) Explain how cancer is a result of uncontrolled cell division.

- Normal cells under control of gene/ signals control /exhibit anchorage dependence (ie attachment to surfaces) and density-dependent inhibition
- Cancer cells do not respond to such signals
- Divide excessively, invading other tissues and eventually killing the organism.

1 mark per point; max [2]

(f) Explain why telomerase inhibitor could form the basis of cancer chemotherapy.

- telomerase allows regeneration of telomeres lost at each division (1m), so coding DNA is not lost
- cells can continue dividing indefinitely (1m)
- loss of telomerase activity would mean death of tumor cells (1m)

1 mark per point; max [2]

(g) Suggest why bacteria do not have telomerase

- Bacteria have circular DNA
- No free ends **max [2]**

Question 2

(a)(i) Label the following structures **[2]**

- 1a: small sub-unit of ribosome (30S), 1b: large sub-unit of ribosome (50S)
- 2:tRNA (aminoacyl-tRNA), 3 (amino acid) or methionine

(a)(ii) Describe the sequence in which the complex shown in Figure 2.1 is assembled. [3]

- Activation of tRNA by aminoacyl-tRNA synthetase to attach amino acid to tRNA
- Small ribosomal sub-unit attaches to mRNA, together with aminoacyl-tRNA carrying methionine;
- At position where anti-codon (on tRNA) is complementary to mRNA codon;
- The large sub-unit then attaches to the complex with the tRNA positioned at the P (peptidyl site)

(a)(iii) Describe the sequences following the formation of the complex shown in Figure 2.1, leading to the formation of a <u>single</u> peptide bond. [3]

- A second aminoacyl-tRNA carrying the second amino acid becomes associated with the complex based on the complementary codon/anti-codon base pairing;
- Occupying the A (aminoacyl) site;
- The large sub-unit of ribosome catalyses the formation of a peptide bond between the two amino acids adjacent to each other;
- Mention of peptidyl transferase (enzymatic portion of large sub-unit) catalysing reaction;
- Further mention of the first aminoacyl-tRNA leaving the complex and translocation of ribosome may be accepted;

(b)(i) Write down the sequence of start codon represented by XXX in Figure 2.1. [1]

- AUG

(b)(ii) Write down the amino acid sequence of the polypeptide being formed. [1]

- Methionine-glutamate-proline-leucine-valine-alanine-glycine

(c) Write down the precise location of a nucleotide change in a specified codon (within the mRNA shown), along with the named type of mutation, that might lead to such a situation described. [1]

- Substitution of G with U at the first nucleotide of 2^{nd} codon.
- Insertion of U before second codon: GAG -> UGA G

[Total: 11m]

<u>Question 3</u>

(a) Explain what is meant by regulatory genes. [1]

- Stretch (sequence of DNA) coding for repressor that binds to the operator, regulating gene expression
- Genes that regulate / control / influence the expression of other genes

(b) Name the three types of enzymes manufactured by each structural gene. [1]

- <u>β-galactosidase</u>, lactose <u>permease</u> and β-galactoside <u>transacetylase</u>;

(c) Explain with reference to Figure 3.1, the control of transcription of the *lac* operon in the presence of lactose. (Reference to the CAP site is not required) [4]

- *LacI* produces repressor that normally binds to operator site near promoter, preventing transcription by RNA polymerase
- Presence of small amounts of allolactose (isomer of lactose) binds to repressor and causes it to change its conformation (note: allolactose is converted from lactose by β-galactosidase in an alternative reaction to hydrolytic one)
- The repressor is no complementarily shaped to operator leaves the operator site.
- RNA polymerase binds to / bound to promoter can now proceed to transcribe the structural genes.
  Note: CRP (cAMP receptor protein) = CAP (catabolite activator protein)

(d) State clearly what AUG in the mRNA strand in Figure 3.1 represents. [1]

- The first mRNA codon for every structural gene, coding for methionine

(e) With reference to Figure 3.2, explain the positive gene control of the *lac* operon. [4]

- When glucose level is low in the bacteria, cAMP will be produced
- cAMP binds to cAMP-activator protein, which binds to CAP site
- Binding of CAP to CAP site speeds up the expression of *lac* operon
- If glucose is absent and lactose present, the CAP site will be bound but repressor does not bind – optimal levels of transcription.
- In the presence of glucose and absence of lactose, cAMP not produced and CAP not bound but repressor binds to operator – no transcription.

- In the presence of both glucose and lactose, repressor does not bind to operator but no cAMP produced and CAP not bound – low levels of transcription
- When glucose and lactose are absent CAP site is not bound and while operator site is bound (no transcription).

(f) With the aid of Figure 3.3, identify and explain the function of regulatory sequences, which are not found in the prokaryotic genome. [4]

- Regulatory sequences are not transcribed;
- Identified as TATA box enabling transcription factors to bind to DNA and recruiting RNA polymerase to the complex;
- Silencers and enhancers in 3a and 3b
- These regulatory sequences (especially enhancers) could be distal or proximal, upstream or downstream
- Activators bind to enhancers and help to speed up transcription by association with the transcription complex.
- If repressors bind to silencer, the expression of the whole gene is stopped.

<u>Question 4</u>

(a) Given that the normal allele is denoted by N and the allele coding for narcolepsy is n, write down the genotypes of the following: [1]

- Individual 2(nn), 4 (Nn), 14 (Nn)

(b) Using specific examples (individuals and their offspring) from the pedigree shown in Figure 4.1, explain why the gene / allele coding for narcolepsy is autosomal and recessive. [2]

- Autosomal: males and females equally likely to inherit the trait (5 males, 4 females)
- Autosomal: transmission from male parent (6) to female offspring (14, 16) as well as male offspring.
- Recessive: Normal parents (3, 4) producing Narcoleptic offspring (8)
- Reference to skipping of generation (individuals 3 and 4)

Any other valid statements and scenario can be accepted.

(c) Describe with elaborations two structural properties of a neuropeptide. [2]

- Not lipid-soluble (or hydrophilic) - inability to traverse the plasma membrane.
- Ability to bind to and activate a receptor on the plasma membrane (complementarily shaped)
- Protein in nature, made up of amino acids forming a polypeptide chain with tertiary level structure

(d) Explain why all three types of molecular aberrations produce similar symptoms. [3]

- All three result in the signal not being sent and/or amplified in the target cells
- First case: no signal molecule (hypocretin) produced.
- Second case: receptor unable to receive signal.
- Third case: Signal does not have a receptor to bind to (no receptor cell)
- Signal transduction could not take place (no cell response).

(e) Give two suggestions of potential methods for the treatment of narcolepsy. [2]

- Drugs: regular injection of drugs
- Gene therapy to produce functional receptor
- Transplant functional cells / stem cells from a compatible donor.

[Total: 10m]

<u>Question 5</u>

(a) With reference to Figure 3.1, outline how the following will link the metabolic pathways between the chloroplast and mitochondrion of this cell.

(i) triose phosphate and hexose phosphate;

in stroma of chloroplast, triose phosphate (TP) is formed as a product of light independent / dark reaction;(1m) TP moves out of chloroplast into the cytosol and is converted to hexose phosphate (1m)
that can enter glycolysis as a respiratory substrate, to be eventually oxidized during aerobic respiration into pyruvate (1m) which enters the mitochondria to undergo the link reaction / to be converted to acetyl CoA(1m)
max [2]

(ii) gaseous exchange.
In granumn of chloroplast, oxygen is formed as a product of photolysis of water during the light dependent/ light reactions (1m); oxygen diffuse out of chloroplast and act as the final electron acceptor in the electron transport chain (ETC) in mitochondrial matrix(1m); CO_2 released during Krebs cycle/ link reaction released into cytosol which diffuse into stroma of chloroplast and used in dark / light independent reaction/ carbon fixation (1m)
max [2]

(b) With reference to **Figure 3.1,**

(i) explain how it is possible for the oxygen concentration to remain constant.
Oxygen released by chloroplast during the light dependent reaction of photophosphorylaton/ photolysis of water is used by mitochondrion as final electron acceptor, compensation point has been reached(1m) Rate of respiration equal to rate of photosynthesis(1m), thus no net change in oxygen concentration.(1m) **max [2]**

(ii) chloroplasts do not have the transporters that allow them to export ATP to the cytosol. How then does the rest of the cell get the ATP it needs for cellular metabolism?
ATP derived from mitochondria / oxidative phosphprylation are supplied to the cell to meet its need(1m) ; triose phosphate from light-independent/ dark reaction will move to cytosol and may eventually be used as respiratory substrate(1m) to drive glycolysis and thus generate ATP.(1m)
max [2]

(c) State the two structural differences between cellulose and starch.

- Cellulose basic units β glucose while starch is composed of α glucose
- Cellulose is composed of straight chains polysaccharide while starch made of polysaccharide chains that are coiled into helices
- Cellulose has hydrogen bonding between adjacent chains while no such hydrogen bonds exist between the polysaccharide chains.
- Cellulose polysaccharide chains are unbranched while in starch (amylopectin), there is branching (glycosidic bond is between carbons 1 and 6)

1 mark per point/ max [2]

(d) Explain how these differences allow them to serve different functions in a plant cell.
Cellulose: long straight chains allow for stacking and formation of hydrogen bonds to form bundles of fibrils to give tensile strength (1m) which is necessary in plant cells to give support and structure (1m).
Starch is coiled and compact to serve as storage material (1m), occupying less space and is insoluble so it does not affect osmotic potential of cell, (1m)

1 mark per point/ max [2]

Question 6

6a) Explain the purpose of having culture A when the aim of the experiment was to investigate the effect of insulin on the cell culture. [1]

- Act as a control to eliminate the possibility of influence of other variables (such as time), instead of insulin causing the difference / to compare glucose changes without insulin added.

(b) Describe the changes in glucose levels observed in both cell cultures. [2]

- Glucose levels in both cell culture media dropped but the level in culture B dropped more than that in culture A
- At the end of 60min, level in culture A was 149 but that in B was around 90mg/dL.

(c) Suggest an explanation for the glucose profile observed in the medium for cell culture A even though insulin is absent. [1]

- Other glucose transporters can still take up glucose at a reduced rate / respiration and other cell processes are still using up glucose

(d) Describe the mechanism of glucose transport for glucose transporter-4. [1]

- Facilitated diffusion: from a region of high to low concentration, requiring the presence of a protein channel

(e) Describe the mechanism of action of the enzymatic portion of the insulin receptor. [2]

- Dimer formation (activated by insulin binding) causes autophosphorylation of each monomer
- ATP is added to tyrosine residue on the receptor;
- Tyrosine kinase phosphorylates tyrosine on intracellular enzymes / proteins to activate them (phosphorylation cascade);

(f) Suggest where glucose transporter-4 is normally located [1]

- Packaged in vesicles ready to be sent to plasma membrane.
- Cytosol (1/2)

(e) Explain the mechanism of cell signal amplification based on Figure 6.2. [2]

- 1 insulin molecule could bind to one receptor that activates several enzymes via phosphorylation;
- Low concentration of signal molecule resulting in many other molecules (in cell response) being produced;
- The enzymes involved (could be reused) activates a much larger number of other other enzymes involved in the phosphorylation cascade;
- Activation of multiple pathways with a single type of signalling molecule;

[Total: 10m]

Question 7

(a) Describe the process of adaptive radiation in the cichlids of Lake Tanganyika.[2]

- Rapid speciation of a single or a few species to fill many ecological niches available.
- Isolated ecosystem for Lake Tanganyika and introduction of a few of such fishes initially (single common ancestor)
- Cichlids quickly speciate to feed on the many different types of food available (some developing into algae-eating, scale-eating, animal-eating etc).

(b) Explain how mouthbrooding behaviour in certain cichlid species might have evolved. [4]

- Variation in breeding habits of cichlids
- Unusual genetic-linked behaviour might have arisen out of chance / mutation
- Mouthbrooding behaviour allows bearer to survive better and/or reproduce more than other cichlids (large percentage of their offspring survive till relatively matured) in the presence of fry-eating predators;
- Inheritable trait genetically passed down to offspring;
- Mechanism of natural selection leading to evolution over many generations;

(c) Account for the changes in frequency of the left-mouthed forms of *Perissodus microlepis* over the period of time surveyed, with reference to Figure 7.1. [4]

- The frequency of left-jawed individual rose and fell on an approximately two-year basis;
- In years 82, 86 and 90-92 it was higher than 0.5 (reverse);
- The form (left or right-jawed) of P. microlepis can act as a type of selection mechanism;
- When a particular form of scale-eater (for example left-jawed) becomes more common, cichlids that are right side dominant would be able to survive better and reproduce more;
- This acts as a selection pressure against left-jawed individuals and they do not survive and reproduce as well as right-jawed individuals. (Thus the alternating cycle)

[Total: 10m]

Section B

Question 8

(a) Compare the ways in which lambda phage and human immunodeficiency virus (HIV) reproduce themselves.

Lambda phage	HIV
Bacteriophage; Infects bacteria	Enveloped animal virus; Infects human T-cells;
Adsorption phase where tail fibers binds to receptors on bacterial cell	Similar phase where glycoprotein spikes (eg gp 120) binds to receptors (eg CD4) on cell surface membrane
Penetration phase: Viral genome (naked) injected into cell through contraction of shaft	Similar phase: Viral genome (contained in capsid) enters cell when viral envelope fuses with cell membrane and capsid injected into cell.
No such phase: Viral genome is naked, no capsid to be digested.	Uncoating: Capsid is disintegrated releasing the viral genome
Prophage formation: ▪ No reverse transcriptase ▪ Viral genome incorporated into host cell genome forming a prophage	Provirus formation • Reverase transcriptase reverse transcribed RNA into DNA • and viral genome incorporated into human genes becoming a provirus
Replicates as part of host cell's genome	
Spontaneous induction occurs resulting in prophage being excised from host genome	Provirus never leaves host genome
Replication: Viral genome replicates, phage components produced by host cell machinery	Similar: Host cell's RNA polymerase transcribes viral DNA into RNA producing viral proteins
Maturation, release and reinfection: Viral components assemble and host cell is lyses, releasing bacteriophages for reinfection.	Similar phases: Viral proteins (capsid, envelope proteins and reverse transcriptase) assemble and viral particles leave host cell via exocytosis, with the host cell membrane forming the viral envelope, to infect other cells.

1 mark per row; max 8

(b) Describe the general structure of animal viruses and explain how these structures are adapted to infecting animal cells. [6]

- Animal viruses composed of phospholipids envelope
- with glycoprotein spikes.

- Enclosed in the viral envelope is the capsid (composed of capsomeres)
- which contains the viral genome (either DNA or RNA)
- glycoprotein spikes are for attachment to host cell membrane
- the envelope (similar in nature to animal cell surface membrane) will fuse with host cell membrane
- allowing entry of the capsid.
- Upon entry, capsid will disintegrate, releasing the viral genome to hijack host cell's metabolic machinery/ be integrated into host cell's genome.

1 mark per point; max 6

(c) Outline how viruses are able to bypass the human defense mechanism to cause diseases. [6]

- Viral proteins and glycoproteins changes the antigenic surface of host cell's cytoplasmic membrane, resulting it host cell being recognized as foreign and destroyed by immune system
- Undergoing antigenic drift, ie changing the glycoprotein spikes on its envelope so the host cell's antibodies cannot bind to the epitopes of viruses.
- Examples are influenza and HIV which can alter their glycoprotein spikes to evade immune system
- Block infected cell from making MHC-1 molecules, thus viral peptide is not displayed on the infected cell and cytotoxic T lymphocytes (CTLs) cannot recognize it and thus cannot be destroyed.
- Eg adenoviruses and Epstein-Barr virus codes for proteins that blocks apoptosis
- Trigger host cells to produce altered MHC-1 molecules that cannot bind viral proteins, therefore not recognized by CTLs.
- Due to presence of altered MCH-1 molecules, natural killer cells are prevented from binding to these infected cells and thus they avoid destruction.

1 mark per point; max 6 **[Total: 20]**

9 (a) Discuss how the membrane structure enables it to perform its function of being selectively permeable. [6]

- Membrane composed of phospholipids bilayer with proteins interspersed
- Proteins are either intrinsic (integral) or extrinsic (peripheral)
- Intrinsic proteins span the membrane width and can serve as channels for movement of lipid insoluble substances.
- Intrinsic proteins can act as ion channels, ligand-gated channels or protein pumps
- Ion channels will facilitated movement of substances across the membrane following laws of simple diffusion
- Ligand-gated channels will undergo conformational change to its structure in the presence of a stimulus, thus allowing movement of substances across the membrane
- While protein pumps, with the aid of ATP, transport ions/ substances across the membrane against concentration gradient
- Creating differences in concentration of substances inside and outside of the cell.
- Through these protein channels/ pumps, the membrane is able to allow substances across it selectively.

1 mark pre point, max 6

(b) Explain how an organism is able to maintain a resting membrane potential of -70mV across the plasma membrane. [8]

- First, on the plasma membrane sodium channels are closed
- Some potassium channels are opened, thus there is a tendency for potassium to leak out of the neurone, thus losing positive charges.
- Second, the sodium/potassium pumps works to transport three sodium ions out of the neurone while taking in 2 potassium ions in
- There is again a net loss of one positive charge within the neurone.
- The membrane potential across the neurone becomes more negative relative to the exterior.

- Finally, together with the presence of negatively charged amino acids within the axoplasm,
- the membrane potential across the plasma membrane becomes even more negative, compared to the extracellular fluid
- 3 factors work to maintain the resting membrane potential at -70mV.

1 mark per point, max 6

(c) Describe the mechanism of synaptic transmission and contrast this with the transmission of a nerve impulse. [6]

Synaptic transmission

- Impulse arrives at synaptic knob, causes deploariztion of pre synaptic membrane
- Ca ions diffuse into synaptic knob, causing the synaptic vesicle to fuse with the pre-synaptic membrane, releasing neurotransmitters into the synaptic cleft.
- Neurotransmitter diffuse across the synaptic cleft, binds to receptors on post-synaptic membrane,
- causing sodium channels on post-synaptic membrane to open, leading to influx of sodium ions thus resulting in depolarization.
- Thus impulse is transmitted across the synapse

1 mark per point, max 4

Compared to transmission of a nerve impulse, synaptic transmission is:

- Slower
- Involved chemical substances to transmit impulse
- Impulses can be summed but nerve transmission is all or none

1 mark per point, max 2

[Total:20]